

Machine Learning Final Project Report

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1. Introduction

Breast cancer is one of the most prevalent malignancies worldwide, representing both a significant clinical and public-health challenge. Early detection is critical: the WHO reports that in high-income countries, age-standardized breast cancer mortality dropped by 40% between the 1980s and 2020, largely due to improvements in early diagnosis and treatment. This progress highlights how critical early health system strengthening is.

Fine Needle Aspiration (FNA) biopsy is a minimally invasive procedure in which a thin needle extracts cells from a suspicious breast mass. Street, Wolberg and Mangasarian (1993) developed an automated system to delineate cell nucleus boundaries from digitized FNA slides, deriving ten nuclear characteristics: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. For each characteristic, three statistics are computed across all nuclei in the image: the mean, the standard error (SE), and the worst (largest) value, yielding exactly 30 numerical features per sample.

Machine learning is a natural fit for this problem. The 30 features capture subtle combinations of measurements that are difficult for human observers to integrate consistently, and automated prediction can serve as a reliable second opinion for pathologists. Notably, the original authors achieved 97% cross-validation accuracy using only 3 of the 30 features, namely mean texture, worst area, and worst smoothness, which motivates investigating which features and methods are most predictive (Street et al., 1993).

This report analyses the Wisconsin Breast Cancer dataset from two angles. First, unsupervised learning is applied to explore whether the data contain natural groupings that align with the true diagnosis, without ever using the labels. Second, a suite of supervised classifiers is trained to predict malignancy directly. The two parts are then explicitly connected: do the clusters discovered correspond to clinical diagnoses? Do unsupervised structures explain where classifiers fail?

2. Problem Formulation

2.1 Unsupervised Question

Is there hidden structure in the 30-dimensional feature space? We use PCA, t-SNE, and clustering algorithms to discover natural groupings in the data, evaluated using the silhouette score, which is a purely internal, label-free metric. As an outlook, we later verify whether the discovered structure corresponds to the true diagnosis.

Unsupervised criteria: Silhouette score ≥ 0.30 , assessed purely from geometry, no labels used in fitting or evaluation.

2.2 Supervised Question

Can we reliably predict whether a tumor is malignant or benign from the 30 nucleus measurements? We train and compare four classifiers, Logistic Regression, Linear SVM, k-Nearest Neighbors, and Random Forest, prioritizing high recall for malignant cases, since a missed cancer (false negative) is far more dangerous than an unnecessary follow-up (false positive).

Success criteria: Recall (primary) ≥ 0.95 on the held-out test set; F1-score (secondary) ≥ 0.93 .

2.3 Why Both Questions Matter

Clinically, if the data cluster naturally by diagnosis without ever observing the labels, it validates that the 30 FNA measurements capture a genuine and coherent biological signal and not just statistical noise. This would suggest that malignant and benign tumors differ systematically at the level of nuclear morphology, which is precisely what FNA-based diagnosis relies on. Methodologically, connecting the unsupervised and supervised results allows us to understand where in feature space classifiers struggle: if misclassified samples fall in regions of geometric ambiguity identified by clustering, it confirms that model errors are not random but reflect genuine overlap between the two classes.

3. Data Understanding & EDA

3.1 Dataset Overview

The Wisconsin Breast Cancer dataset was loaded directly from scikit-learn. It contains 569 samples and 30 continuous numerical features per sample, derived from digitized FNA cell-nucleus images, and a binary target label: Malignant ($M = 1$) or Benign ($B = 0$).

The target variable was recoded so that 1 = Malignant and 0 = Benign, making malignancy the positive class (clinically the important one). No missing values were found across any of the 30 features.

The 30 features are organized as ten nuclear measurements, namely: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. Each is summarized with three statistics: mean, standard error (SE), and worst (largest) value.

3.2 Class Distribution

The dataset is moderately imbalanced: 357 samples (62.7%) are Benign and 212 (37.3%) are Malignant. A naive classifier always predicting “benign” would achieve 62.7% accuracy but zero recall for the clinically critical malignant class. This imbalance motivates using F1-score as the secondary metric alongside recall.

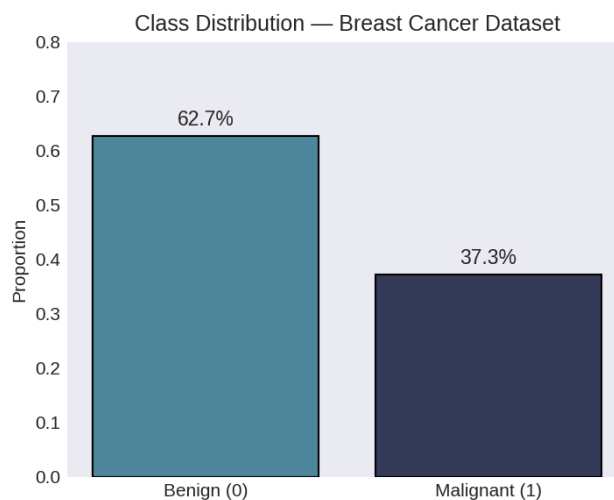


Figure 1. Class distribution of the dataset (Benign = 62.7%, Malignant = 37.3%).

3.3 Feature Distributions by Class

The histograms for the ten mean features are shown in Figure 2, where the x-axis represents the feature value and the y-axis the number of samples at that value, shown separately for benign and malignant cases. The SE and worst features are not plotted separately as they largely mirror the same patterns, since they are derived from the same underlying nuclear measurements, however, this redundancy is confirmed quantitatively in the correlation analysis (Section 3.4).

Among the mean features, clear separation between benign and malignant classes is visible for size-related measurements: malignant tumors consistently show larger mean radius, perimeter, area, concavity, and concave points. This is biologically meaningful, since we can already assume that malignant cells tend to be larger and more irregularly shaped, with greater internal concavity reflecting the uncontrolled growth patterns characteristic of cancerous tissue. Mean fractal dimension, however, shows almost no separation between the two classes. Fractal dimension measures the complexity of the nucleus boundary at a fine scale, so the fact that it does not discriminate suggests that while malignant nuclei are larger and more concave, their boundary complexity at the microscopic level is not systematically different from benign nuclei. This feature is therefore unlikely to contribute meaningfully to classification.

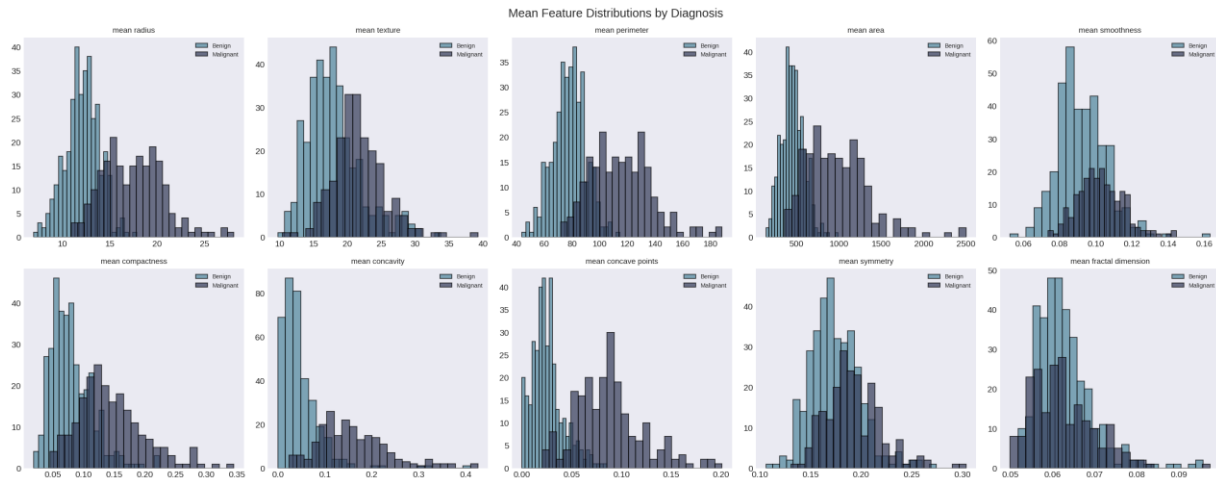


Figure 2. Mean feature distributions by diagnosis class.

The boxplots in Figure 3 show all ten mean features side by side. Mean radius, mean perimeter, and mean area show the strongest separation, the malignant median being consistently and substantially higher, with a wider interquartile range, reflecting both the larger size and greater variability of malignant nuclei. Mean concavity and mean concave points also discriminate well, with malignant tumors showing notably higher values and more spread. Mean compactness follows a similar pattern, while mean texture shows moderate separation. Mean symmetry and mean smoothness show the least separation among the clearly discriminative features, although their interquartile ranges overlap considerably. Mean fractal dimension shows the least separation of all, with the two distributions almost entirely overlapping, further confirming that fine boundary complexity does not distinguish the two classes and is unlikely to contribute meaningfully to classification.

Several features (notably mean texture, mean compactness, mean concavity, mean concave points, and mean fractal dimension) display outliers in both classes, predominantly high-value extremes in the benign group. These were retained in the analysis as they likely reflect genuine biological variation rather than measurement error, and their influence was mitigated by standardizing all features prior to modeling.

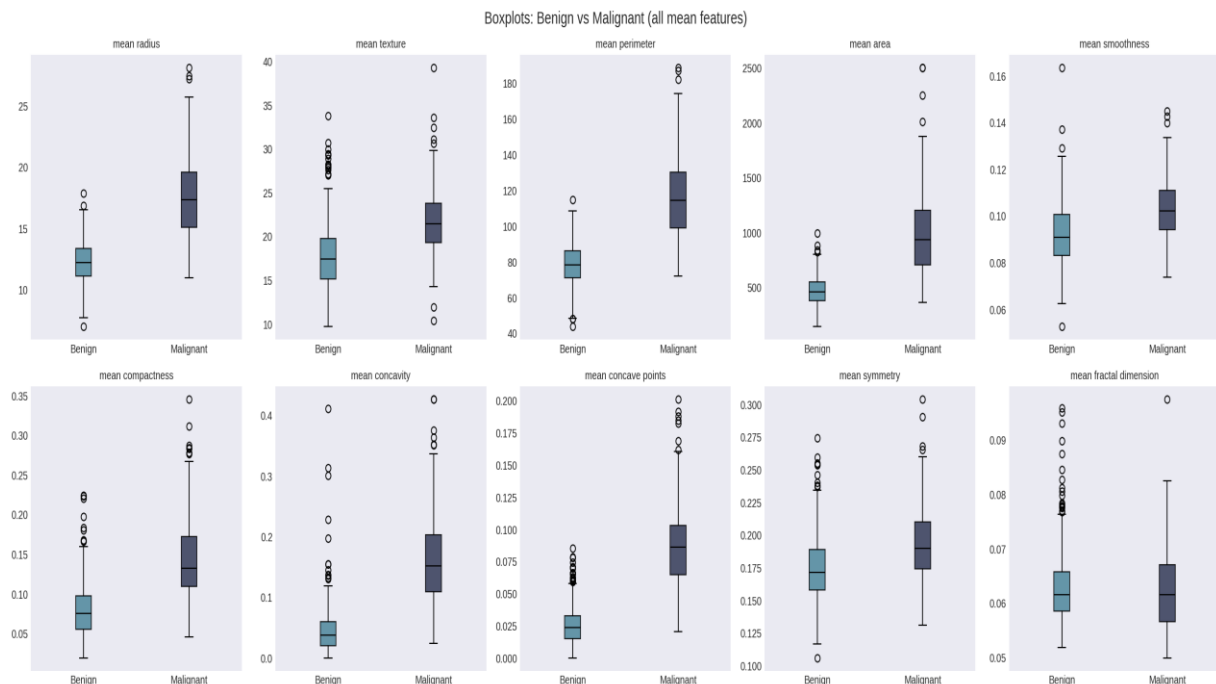


Figure 3. Boxplots of the mean features comparing Benign vs. Malignant classes.

3.4 Correlation Analysis

The full 30×30 correlation heatmap in Figure 4 shows the pairwise correlations between all features, where purple indicates strong positive correlation and blue indicates negative correlation. The most striking pattern is the substantial redundancy across the feature space: the mean, SE, and worst triplicates for each nuclear measurement are highly correlated with each other ($r > 0.9$ for radius, perimeter, and area), which is expected since they are computed from the same underlying nuclear boundaries. The size-related groups are also strongly correlated with each other across all three statistics, as they all fundamentally capture cell size from slightly different angles.

Negative correlations are present but weak, appearing most notably between fractal dimension and the size-related features. This is consistent with the previous EDA findings that fractal dimension carries little discriminative information. It measures a fundamentally different property of the nucleus boundary than size or concavity.

This redundancy has a direct practical implication, which is that PCA should be able to compress the 30 features into a much smaller number of components without significant information loss. As confirmed in Section 5.1, just 7 principal components capture 90% of the total variance, meaning the effective dimensionality of this dataset is far lower than the nominal 30 features suggest.

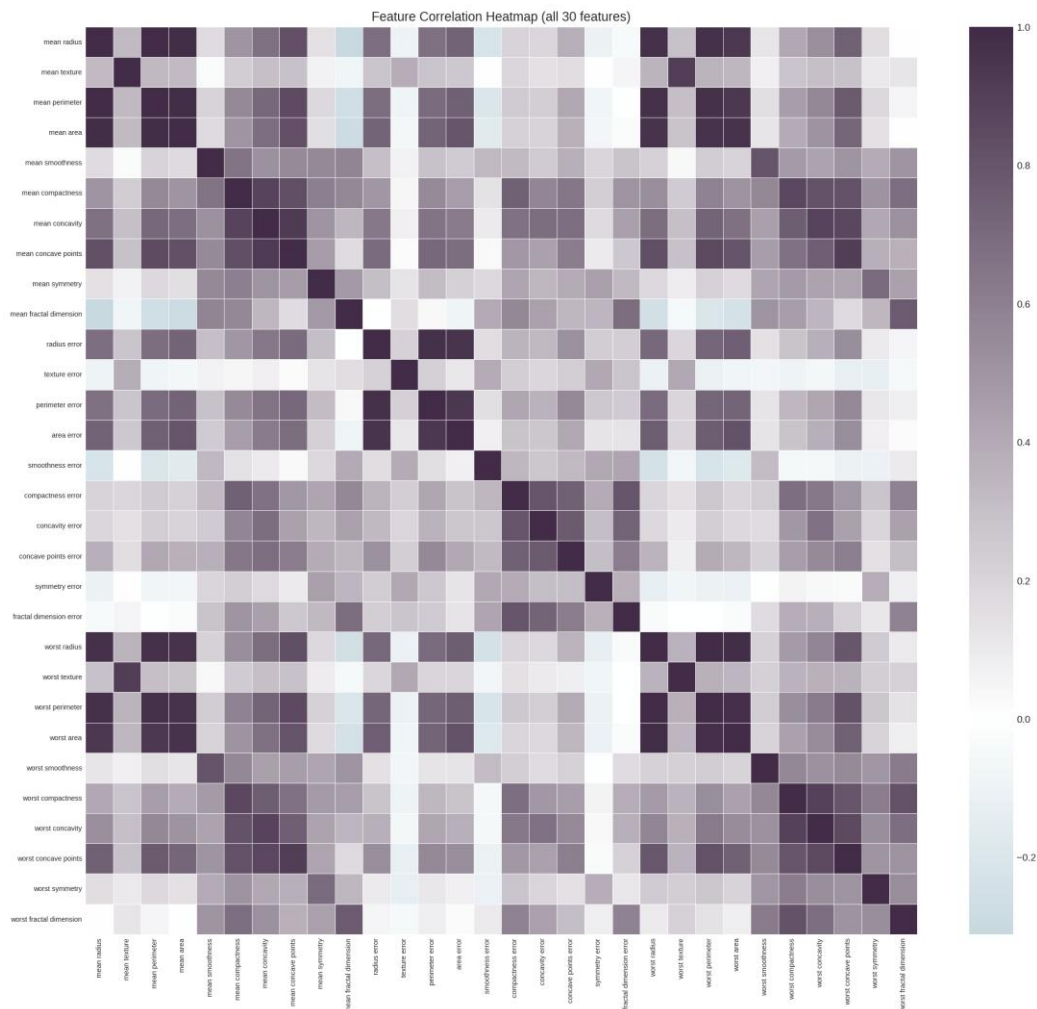


Figure 4. Feature correlation heatmap (all 30 features). Purple indicates strong positive correlation; blue indicates negative correlation.

4. Preprocessing

4.1 Train/Test Split

The dataset was split into a training set (80%, 455 samples) and a test set (20%, 114 samples) using stratified sampling, preserving the 63/37 class ratio in both splits.

Strictly speaking, unsupervised learning does not require a train/test split, since clustering and dimensionality reduction have no concept of generalization to unseen labels. However, the split is applied consistently throughout the report for two reasons: first, PCA fitted on the training data is reused as a preprocessing step in the supervised section, so consistency requires it to be fitted on the same training partition; second, the label-based verification in the unsupervised outlook (Section 5.5) uses the same split to ensure no information from the test set influences any learned transformation.

4.2 Standardization

All 30 features are continuous numerical variables with very different scales. Standardization using StandardScaler (z-score normalization) is essential:

- PCA is variance-based, so large-scale features would otherwise dominate the principal components,
- kNN is distance-based, so unstandardized features bias Euclidean distances toward high-magnitude dimensions,
- and SVM and Logistic Regression assume comparable feature scales for regularization and margin computation.

4.3 No Categorical Features

All 30 features are numerical, so no encoding is required, simplifying the pipeline considerably compared to datasets with mixed feature types.

4.4 Leakage Prevention

All preprocessing and training steps, including StandardScaler and PCA, are always fitted on training data only and applied to test data, ensuring no information from the test set influences any learned transformation.

5. Unsupervised Learning

All unsupervised models are fitted and evaluated without using class labels. The silhouette score is the sole evaluation metric in this section, which is a purely internal, geometry-based metric. Class labels are shown in some visualizations for interpretability, and a brief label-based verification appears at the end as an outlook only.

5.1 PCA — Principal Component Analysis

PCA was fitted on the training data only. The scree plot and cumulative explained variance analysis in Figure 5 show that just 7 principal components capture 90% of the total variance, and 10 components reach 95%. This confirms the high redundancy observed in the correlation heatmap: the original 30 features can be compressed by more than a factor of 4 without meaningful information loss.

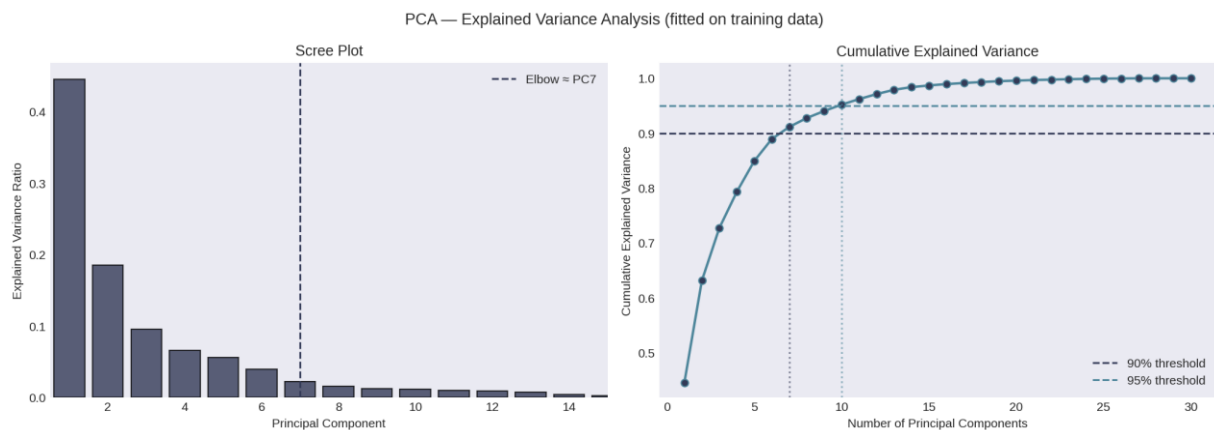


Figure 5. PCA explained variance — scree plot (left) and cumulative variance (right). The purple dashed line marks the 90% threshold at PC7; the blue line marks 95% at PC10.

The loading analysis in Figure 6 reveals what each principal component represents:

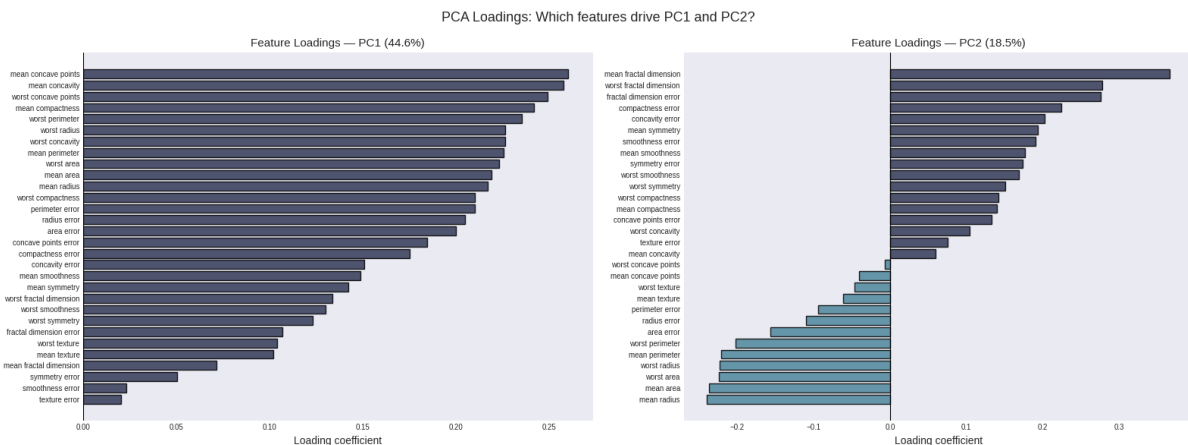


Figure 6. Feature loadings for PC1 (left) and PC2 (right). Features with larger absolute loadings drive the respective component.

PC1 (44.6% of variance) is driven primarily by size and shape measurements: mean concave points, mean concavity, worst concave points, mean compactness, and worst perimeter all show the highest positive loadings, with coefficients around 0.20–0.26. This means PC1 essentially captures how large, irregular, and concave a nucleus is, which are precisely the features that best separated the two classes in the EDA.

PC2 (18.5% of variance) captures a different dimension of variation: driven positively by fractal dimension and texture features, and negatively by size-related features such as mean radius, worst area, and worst perimeter. This suggests PC2 distinguishes samples that score high on boundary complexity

and texture but low on size, displaying a secondary axis of biological variation largely independent of malignancy, consistent with the EDA finding that fractal dimension and texture are less discriminative.

The 2D PCA projection in Figure 7, colored by K-Means cluster assignment, reveals a clear separation between two groups along PC1, confirming that the first principal component acts as a near-complete separating axis for the two biological groups. The K-Means clustering itself is discussed in detail in Section 5.3.1.



Figure 7. PCA 2D projection colored by K-Means clusters ($k=2$, no labels used). Centroids are shown as black crosses.

5.2 t-SNE Visualization

t-SNE is used here purely for visualization. Unlike PCA, it has no transform() method. Once fitted, it cannot generate coordinates for new, unseen data points, making it impossible to use as a preprocessing step for a classifier. All t-SNE fits use only the training data, scaled with the training-fitted scaler.

A key hyperparameter in t-SNE is perplexity, which controls the balance between preserving local and global structure in the 2D projection: low perplexity focuses on very local neighborhoods, while high perplexity considers broader structure. A perplexity sensitivity analysis was conducted at values 5, 20, 30, and 40. The silhouette score of K-Means clusters in t-SNE space was used as a relative proxy for perplexity selection and perplexity = 40 was selected as it produced the highest silhouette score (0.5521).

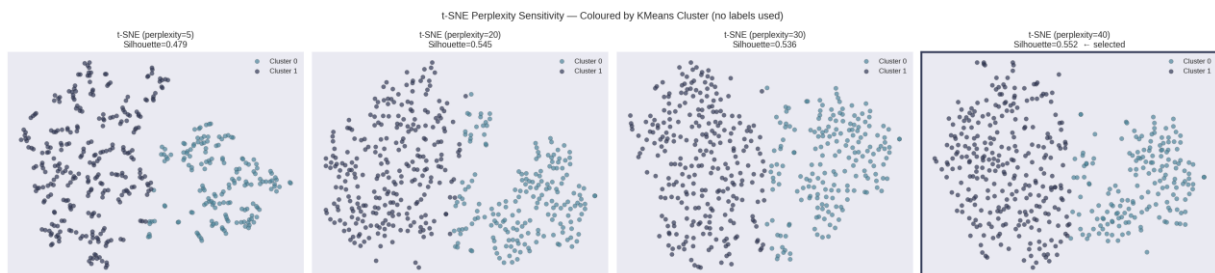


Figure 8. t-SNE perplexity sensitivity analysis — K-Means clusters at four perplexity values. The selected perplexity (40) is highlighted with a colored border.

It is important to note that these silhouette scores are used only as a relative proxy for this selection and are not directly comparable to silhouette scores in the original feature space, since t-SNE distorts global distances by design.

5.3 Clustering

5.3.1 K-Means

K-Means is a clustering algorithm that partitions the data into k groups by iteratively assigning each sample to its nearest centroid and updating the centroids until convergence.

The elbow plot and silhouette analysis in Figure 9 both confirm that $k = 2$ is the natural partition for this dataset. The elbow occurs sharply at $k = 2$, and the silhouette score is highest at $k = 2$ (0.3432), dropping substantially for all higher values of k . A silhouette score of 0.3432 indicates moderate but meaningful cluster separation, suggesting the two groups are distinct enough to be reliably identified, but with some overlap in the boundary region, which is consistent with the class overlap observed in the EDA.

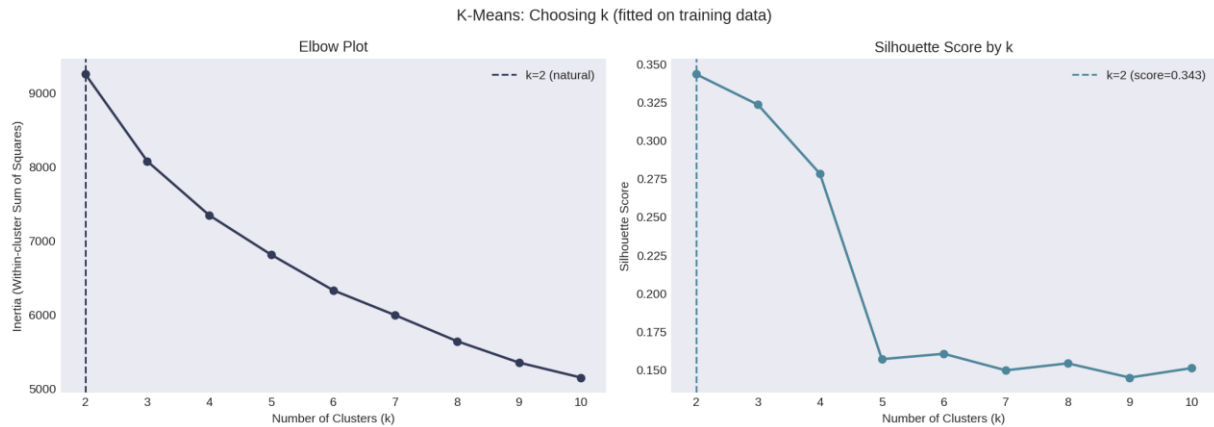


Figure 9. K-Means model selection — elbow plot (left) and silhouette score by k (right). Both criteria confirm $k=2$ as the natural partition.

K-Means ($k = 2$) was fitted on the full 30D data and then visualized in PCA 2D and t-SNE 2D space in Figure 10. Both projections confirm clear separation between the two clusters. In the PCA projection, the separation is primarily along PC1, consistent with the loading analysis showing that PC1 captures the size and shape features that most distinguish the two biological groups. The t-SNE projection shows two broadly separated regions, though with more mixing at the boundary, ultimately reflecting the same overlap zone visible in PCA space.

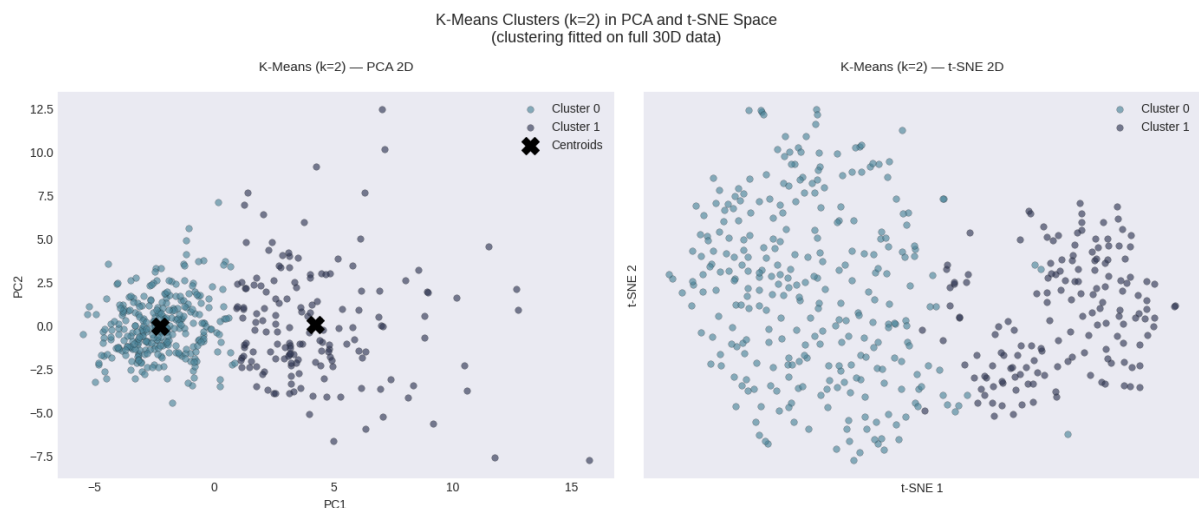


Figure 10. K-Means ($k=2$) clusters in PCA 2D (left) and t-SNE 2D (right).

5.3.2 Gaussian Mixture Models (GMM)

GMM is a probabilistic clustering method that models the data as a mixture of Gaussian distributions. Unlike K-Means, which assigns each point to exactly one cluster, GMM produces soft assignments, meaning each point gets a probability of belonging to each component. This is particularly valuable in

a medical setting where a sample may be biologically ambiguous and does not clearly belong to either group.

GMM pairs especially well with PCA: PCA components are uncorrelated by construction, directly satisfying the GMM covariance assumption. Fitting Gaussians to uncorrelated components is statistically more efficient and requires fewer parameters than fitting to the original correlated 30D space.

BIC (Bayesian Information Criterion) and AIC (Akaike Information Criterion) are model selection criteria that balance how well a model fits the data against how complex it is, since both penalize adding more components if the improvement in fit is not justified, therefore lower values indicate a better model. BIC penalizes complexity more heavily than AIC, reducing the risk of selecting too many components. As shown in Figure 11, BIC reaches its minimum at $n = 2$, confirming that the data naturally support two Gaussian components. AIC continues to decrease beyond $n = 2$, but this is expected, since AIC tends to favor more complex models, making BIC the more reliable criterion here.



Figure 11. GMM model selection — BIC and AIC scores by number of components.

The visualization in Figure 12 shows hard cluster assignments (top row) and soft membership probabilities (bottom row) in PCA 2D and t-SNE 2D. The hard assignments closely mirror the K-Means result, confirming the robustness of the two-cluster structure. More informatively, the soft probability plots reveal a gradient of certainty: samples in the core of each cluster receive probabilities close to 0 or 1, while samples near the boundary between the two groups receive probabilities close to 0.5. These boundary samples are geometrically ambiguous, since they sit in the overlap zone between the two Gaussian components, meaning the model is genuinely uncertain about their assignment. As will be shown in Section 7.4, these are precisely the samples where supervised classifiers are most likely to make errors.

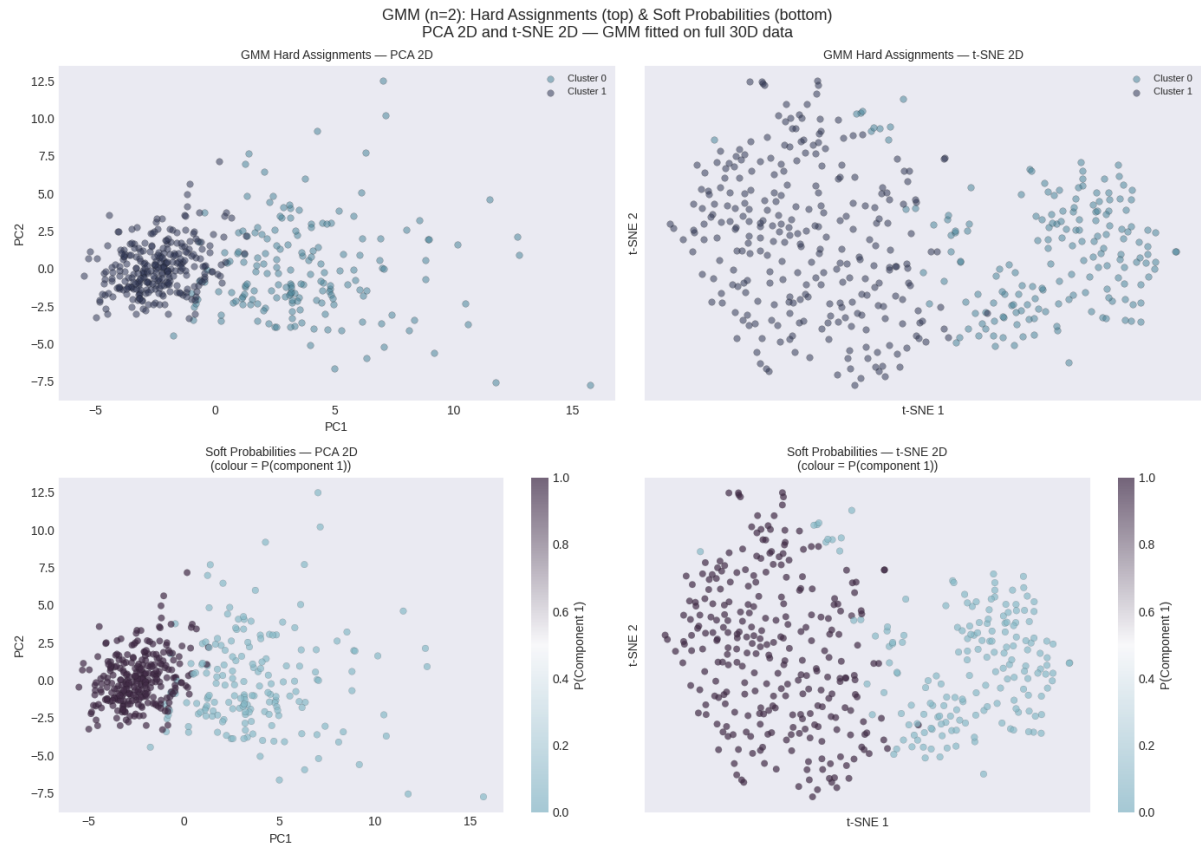


Figure 12. GMM ($n=2$) hard cluster assignments (top) and soft membership probabilities (bottom) in PCA 2D and t-SNE 2D.

5.3.3 Hierarchical Clustering

Hierarchical clustering builds a tree of nested clusters by iteratively merging the most similar samples, without requiring a predetermined number of clusters. The final number of clusters is chosen by cutting the tree at a chosen distance level.

The choice of linkage criterion determines how the distance between clusters is measured. While single, complete and average linkage were covered in class, Ward linkage was selected here instead after testing all four criteria. Ward linkage merges the two clusters that result in the smallest increase in total within-cluster variance at each step, which tends to produce more compact and similarly sized clusters than single, complete, or average linkage. This makes it more appropriate for this dataset where the two biological groups are expected to be of comparable size and spread, and it produced the most interpretable dendrogram structure among the four options tested.

The dendrogram in Figure 13, computed on a 200-sample subsample for visual clarity, shows a clean two-cluster structure with a large gap between the final merge and the previous ones. This large gap at the top of the tree confirms that $k = 2$ is the natural partition and merging beyond this point would combine two very dissimilar groups.

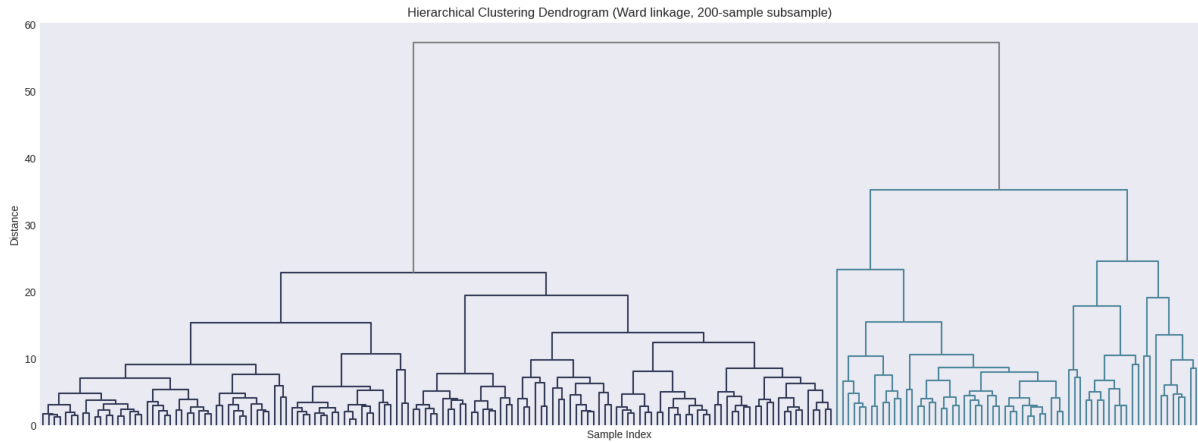


Figure 13. Hierarchical clustering dendrogram (Ward linkage, 200-sample subsample).

Figure 14 shows the resulting clusters in PCA 2D and t-SNE 2D space. The separation is consistent with K-Means and GMM, though the silhouette score (0.2886) is slightly lower than K-Means (0.3432), suggesting that Ward linkage produces slightly less compact clusters than expected on this dataset.

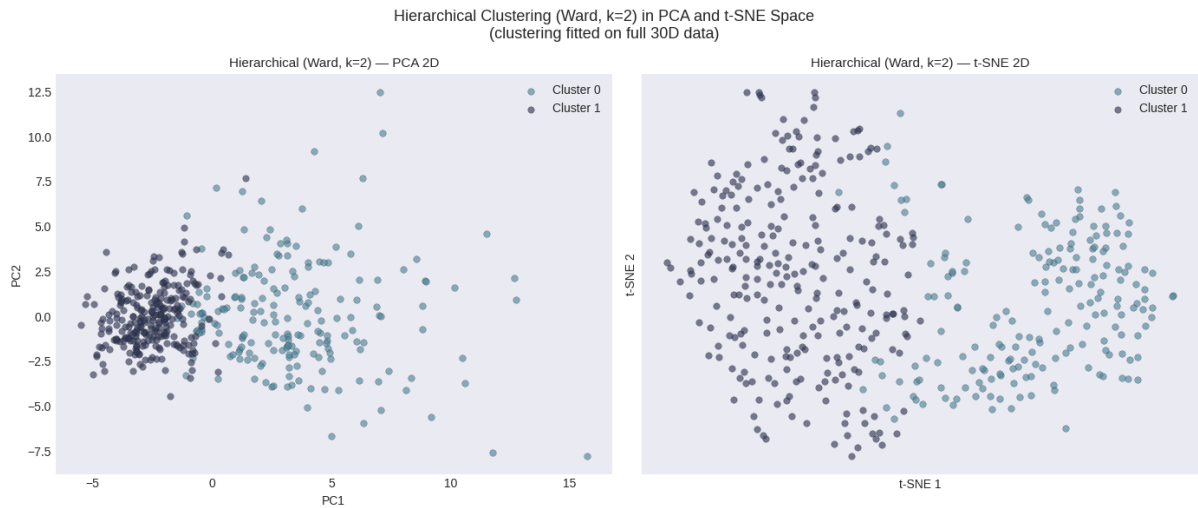


Figure 14. Hierarchical (Ward, $k=2$) clusters in PCA 2D (left) and t-SNE 2D (right).

5.4 Unsupervised summary & comparison

Table 1 and Figure 15 summarize the silhouette scores for all three clustering methods across all three representations. Several important observations emerge.

Method	Baseline (30D)	PCA 2D	t-SNE 2D
K-Means	0.3432	0.5092	0.5521
GMM	0.3142	0.4736	0.5532
Hierarchical	0.2886	0.4821	0.5028

Table 1. Silhouette scores for all clustering methods across representations (label-free).

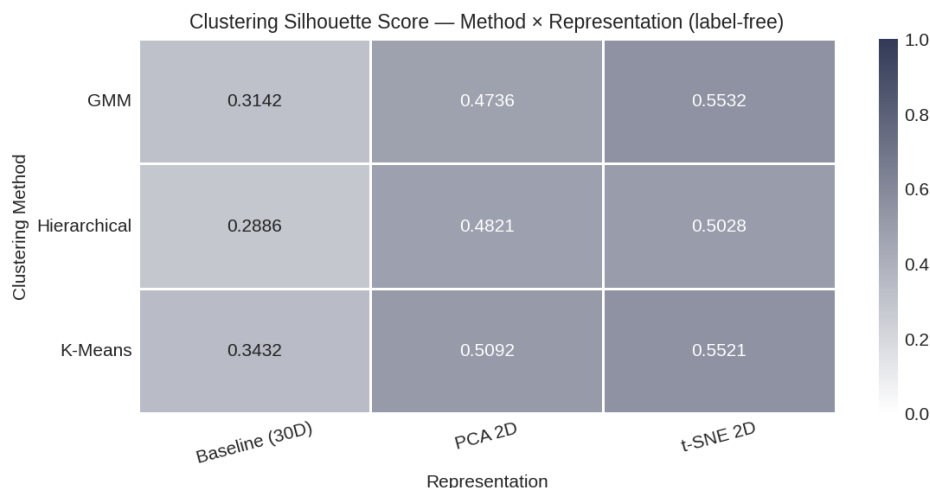


Figure 15. Master clustering comparison — silhouette score heatmap across all methods and representations (label-free).

Two out of the three methods meet the success criterion of silhouette ≥ 0.30 on the baseline 30D data, confirming that genuine cluster structure exists in the original feature space. K-Means achieves the highest baseline score (0.3432), followed by GMM (0.3142) and Hierarchical (0.2886).

Silhouette scores are consistently higher in PCA 2D and t-SNE 2D than in the baseline 30D space. However, these higher scores should not be interpreted as better clustering, since they mainly reflect the properties of the embedding rather than the original data structure. t-SNE explicitly compresses points into tight groups and pushes groups apart, so silhouette scores in t-SNE space are artificially inflated and not directly comparable to baseline scores. PCA 2D is a fairer intermediate, as it is a linear distance-preserving projection. The baseline 30D silhouette therefore remains the most honest measure of cluster quality.

In terms of method-representation pairing, K-Means works best with the baseline 30D data or PCA, since it relies on Euclidean distances which are meaningful in both spaces. GMM is most appropriately paired with PCA, since PCA components are uncorrelated by construction, directly satisfying the Gaussian covariance assumption. Hierarchical clustering with Ward linkage also pairs well with PCA representations, since Ward linkage is a variance-minimization criterion that requires meaningful Euclidean distances. t-SNE is not recommended as a clustering input, only as a visualization tool, since its distances are not preserved and the embedding cannot be applied to unseen data points.

Among the three methods, K-Means is the recommended choice for this dataset: it achieves the highest baseline and PCA silhouette, is computationally efficient, and its hard assignments are straightforward to interpret. GMM is the recommended choice when uncertainty quantification matters, although Hierarchical clustering achieves a slightly higher silhouette on PCA 2D (0.4821 vs. 0.4736), GMM is preferred over Hierarchical because its probabilistic soft assignments identify the geometrically ambiguous samples that are hardest to classify, which proves particularly valuable in the connection with supervised learning in Section 7.4. This additional information about uncertainty is something neither K-Means nor Hierarchical clustering can provide.

5.5 Outlook: Label-Based Verification

As an outlook, the true class labels are revealed to verify whether the internal cluster structure corresponds to clinical reality. This label-based verification was not part of the primary unsupervised evaluation and labels were never used in fitting or silhouette computation.

Here, three metrics are used: ARI (Adjusted Rand Index), which measures overall agreement between cluster assignments and true labels corrected for chance; recall, which measures how well the malignant class is identified; and F1, which balances precision and recall and we mainly implement it since the dataset is slightly imbalanced.

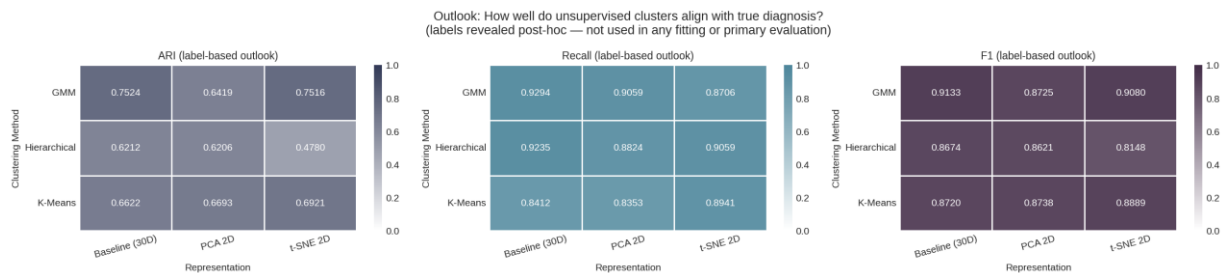


Figure 16. Outlook — label-based verification: ARI, Recall, and F1 heatmaps for all clustering methods and representations. Labels were revealed post-hoc only.

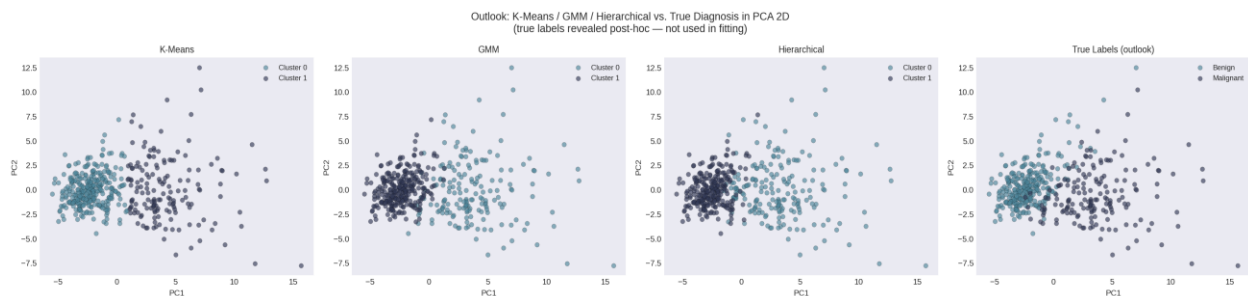


Figure 17. Outlook — K-Means, GMM, and Hierarchical cluster assignments vs. true diagnosis in PCA 2D. The rightmost panel shows the ground truth.

The results are remarkably strong. GMM achieves the highest ARI on the baseline 30D data (0.7524), meaning its cluster assignments agree strongly with the true diagnosis even without ever observing the labels. K-Means and Hierarchical also show solid baseline ARI values (0.6622 and 0.6212 respectively), confirming that all three methods recover a structure that is genuinely aligned with clinical reality.

Recall values are particularly impressive across all methods. GMM achieves 0.9294 on baseline 30D, meaning it correctly identifies over 93% of malignant cases purely from geometric structure, with no label information. Hierarchical achieves 0.9235 and K-Means 0.8412 on the same representation. These are remarkable results for an entirely unsupervised approach and directly confirm that the 30 FNA measurements capture a coherent biological signal that distinguishes malignant from benign tumors at the level of nuclear morphology.

The PCA 2D projection in Figure 17 visually confirms this alignment: the cluster boundaries discovered by all three methods closely match the true diagnosis boundary shown in the rightmost panel, with mismatches concentrated in the overlap zone along PC1, which is precisely the region of geometric ambiguity identified throughout this section.

This strong unsupervised alignment motivates the supervised section: if the data contain such clear structure that even label-free methods can recover the diagnosis with high recall, supervised classifiers with access to the true labels should be able to achieve even stronger performance.

6. Supervised Learning

6.1 Task Definition & Evaluation Strategy

The dataset contains 357 benign and 212 malignant cases, corresponding to a 63/37 class split. This imbalance makes accuracy a misleading metric, since a naive classifier that predicts benign for every sample would achieve 63% accuracy without learning anything meaningful about the data.

In a cancer diagnosis setting, the two types of error carry very different consequences. A false negative occurs when the model predicts benign, but the tumor is actually malignant, meaning a cancer goes undetected and potentially untreated. A false positive, on the other hand, means predicting malignant when the tumor is actually benign, which only leads to further testing and anxiety but is far less dangerous. This asymmetry means that recall for the malignant class is the most clinically important metric, and models that miss malignant cases should be penalized heavily.

However, recall alone is still insufficient. A classifier that predicts malignant for every sample achieves perfect recall but is completely clinically useless, as it would flag every benign case as cancerous. What is needed is a metric that rewards models that are both sensitive to malignant cases and precise enough to avoid excessive false alarms. F1-score achieves this by combining precision and recall through the harmonic mean, which unlike a regular average, penalizes models that are strong on one metric but weak on the other, pushing models toward a genuinely balanced performance.

Considering the above, recall is used as the primary metric throughout this section, given the cost asymmetry of the problem, with F1-score tracked as a secondary metric to ensure precision does not deteriorate excessively.

6.2 Logistic Regression

Logistic Regression assigns a weight to each feature and combines them linearly, then applies a sigmoid function to convert the result into a probability between 0 and 1. A threshold of 0.5 is used to determine the final class assignment.

The regularization parameter C controls how strongly the model is penalized for large weights. A lower C keeps weights small and prevents overfitting. C was tuned via 5-fold stratified cross-validation using GridSearchCV with recall as the scoring metric. The best C was 0.034, which is a relatively low value, indicating that strong regularization was beneficial, keeping the model weights small and preventing overfitting on the training data.

Metric	Value
Train Accuracy	0.9780
Test Accuracy	0.9825
Precision	1.0000
Recall	0.9524
F1-Score	0.9756
AUC	0.9980

Table 2. Logistic Regression test set performance metrics.

The results are strong across all metrics. Most remarkably, precision = 1.000, meaning every single case flagged as malignant truly was malignant, meaning the model produced zero false alarms across all 114 test samples. This is a clinically significant result, though it should be interpreted with some caution since it may be partly a product of the specific test split rather than a guaranteed property of the model. Recall = 0.9524 means that 40 out of 42 malignant cases were correctly identified, with only 2 missed,

which is a strong result given the clinical cost of false negatives. The train-test accuracy gap is negligible (0.9780 vs. 0.9825), confirming that the model generalizes well, and it is not overfitting.



Figure 18. Logistic Regression confusion matrix on the test set.

The feature coefficient plot in Figure 19 reveals which features drive the model's decisions. The top predictors by absolute coefficient value are worst texture, worst radius, worst perimeter, worst area, and mean concave points, all with coefficients around 0.30–0.40. This is largely consistent with the EDA and PCA findings: size-related and shape-related features dominate, while fractal dimension does not appear among the top 10, confirming its low discriminative value. Interestingly, worst texture appears as the single most influential feature, which differs slightly from the EDA where texture showed only moderate separation, suggesting that in combination with other features, texture carries more predictive weight than its distribution alone would suggest.

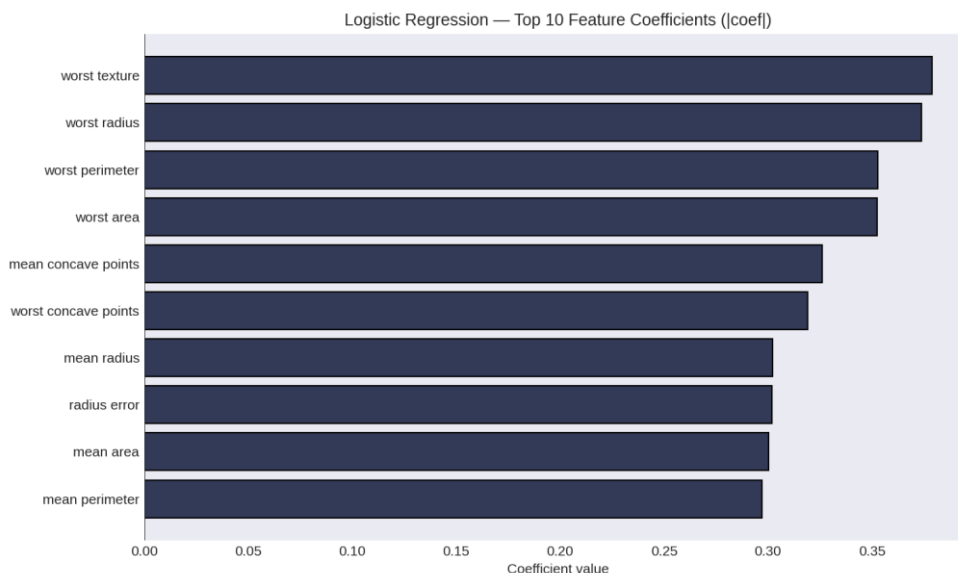


Figure 19. Logistic Regression — top 10 feature coefficients (by absolute value). Positive coefficients are associated with the malignant class.

6.3 Linear SVM

Linear SVM finds a decision boundary that separates the two classes with the largest possible margin. Unlike Logistic Regression, which minimizes a probabilistic loss across all training points, SVM focuses only on the support vectors, the points closest to the boundary, and maximizes the geometric

margin between them. C was tuned using the same 5-fold stratified cross-validation procedure as Logistic Regression, with recall as the scoring metric.

Metric	Value
Train Accuracy	0.9956
Test Accuracy	0.9386
Precision	0.9487
Recall	0.8810
F1-Score	0.9136
AUC	0.9679

Table 3. Linear SVM test set performance metrics.

The results are notably weaker than Logistic Regression across all metrics. Recall = 0.8810 means 5 malignant cases were missed, compared to only 2 for Logistic Regression, which is a meaningful difference given the clinical cost of false negatives. Precision = 0.9487 and F1 = 0.9136 are both lower than LR's corresponding values. Most notably, the train-test accuracy gap is substantial (0.9956 vs. 0.9386), indicating overfitting.

This can be explained by the best C found: 29.76, which is a very high value, meaning almost no regularization was applied. With such a high C , the model is allowed to fit the training data very closely, prioritizing correct classification of every training point over finding a generalizable boundary. This contrasts sharply with Logistic Regression, where the best C was 0.034, indicating that strong regularization was beneficial and kept the model from overfitting. The choice to optimize hyperparameters using recall as the scoring metric further amplifies this effect: GridSearchCV is incentivized to select values of C that maximize sensitivity on training folds, systematically favoring low-regularization solutions even at the cost of generalization. Moreover, Logistic Regression's probabilistic loss considers all training points and produces a smoother, better-calibrated decision boundary, while SVM's focus on support vectors alone makes it more sensitive to the specific samples near the boundary, which can hurt generalization on small datasets.

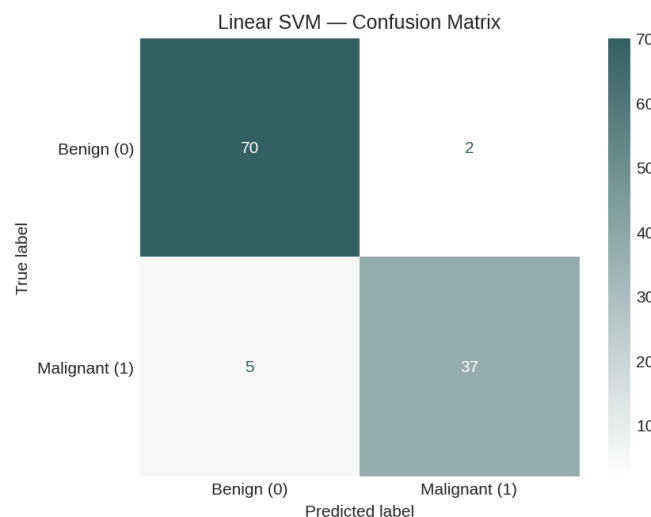


Figure 20. Linear SVM confusion matrix on the test set.

6.4 k-Nearest Neighbors

k-Nearest Neighbors predicts the class of a new sample by looking at the k most similar samples in the training data and taking a majority vote. Similarity is measured by Euclidean distance in feature space, which is why standardization is essential here and without it, features with larger numerical ranges

would dominate the distance calculation regardless of their actual predictive relevance. The number of neighbors k was tuned using recall as the scoring metric, over odd values from 1 to 21 to avoid tied votes. The upper bound of 21 follows the common practice of taking the square root of the number of training samples: $\sqrt{455} \approx 21.3$.

The bias-variance plot in Figure 21 shows CV recall as a function of k . The best k selected was $k=3$, achieving a CV recall of 0.935. As k increases beyond 3, recall decreases relatively steadily, reflecting increasing bias as the model averages over too many neighbors and loses sensitivity to local malignant clusters.

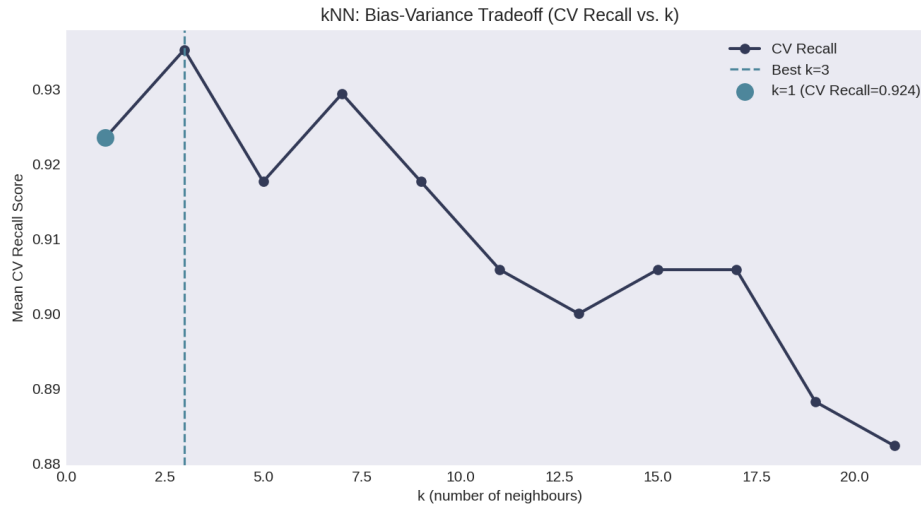


Figure 21. k NN bias-variance tradeoff — cross-validation Recall by k . $k=1$ (high variance, low bias) is shown as a reference point.

Metric	Value
Train Accuracy	0.9846
Test Accuracy	0.9386
Precision	0.9730
Recall	0.8571
F1-Score	0.9114
AUC	0.9825

Table 4. k NN (best k) test set performance metrics.

The test results show a similar pattern to Linear SVM: a notable train-test gap (0.9846 vs. 0.9386) suggests some overfitting even at $k=3$. Recall = 0.8571 means 6 malignant cases were missed, giving the worst recall of all models tested so far. Precision is high at 0.9730, but given that recall is the primary metric here, k NN is not the recommended model for this task. The relatively poor recall is consistent with the curse of dimensionality: in high-dimensional spaces, all points tend to become equidistant, making neighborhood structure less reliable and reducing the model's ability to correctly identify the malignant minority class. It is important to note that standardization does not resolve this issue. While it eliminates scale differences between features, all 30 dimensions remain present, and the fundamental problem of distances becoming increasingly similar in high-dimensional spaces persists regardless of scaling.

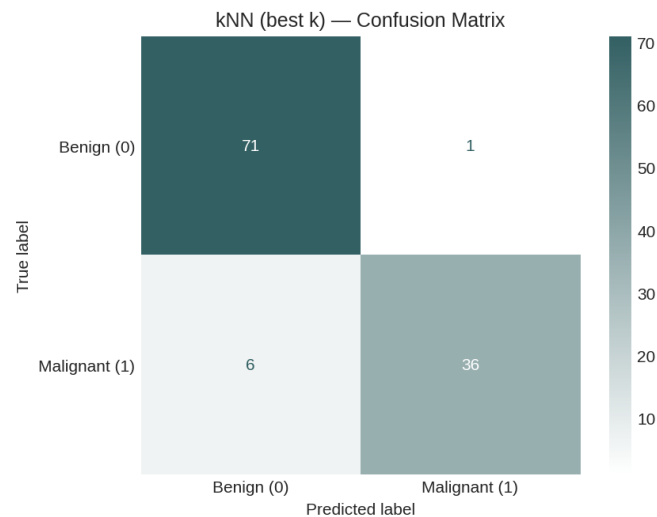


Figure 22. kNN (best k) confusion matrix on the test set.

6.5 Random Forest

Random Forest is an ensemble method that builds multiple decision trees on random bootstrapped subsets of the training data, using a random subset of features at each split. The final prediction is determined by majority vote across all trees. This combination of bootstrapping and feature randomness acts as a built-in regularization mechanism, reducing variance compared to a single decision tree. The hyperparameters tuned were the number of trees (`n_estimators`), maximum depth (`max_depth`), and minimum samples per split (`min_samples_split`).

The best parameters found were `n_estimators=100`, `max_depth=None`, and `min_samples_split=2`. Setting `max_depth` to `None` means each tree is allowed to grow until it perfectly classifies every training sample, which is why training accuracy reaches 1.000. A single tree doing this would be pure memorization and would generalize poorly. However, Random Forest builds 100 different trees, each trained on a different random subset of the data and using different random subsets of features at each split. Because each tree sees a slightly different version of the data, no single training sample dominates all trees, and the majority vote across 100 trees smooths out individual errors. This is why the model still generalizes well to the test set despite the perfect training accuracy. The train-test gap (1.000 vs. 0.9737) is therefore acceptable and expected for Random Forest.

Metric	Value
Train Accuracy	1.0000
Test Accuracy	0.9737
Precision	1.0000
Recall	0.9286
F1-Score	0.9630
AUC	0.9929

Table 5. Random Forest test set performance metrics.

The results are strong. Precision = 1.000 means zero false alarms again. However, recall = 0.9286 means 3 malignant cases were missed, which is better than SVM and kNN but slightly worse than Logistic Regression.

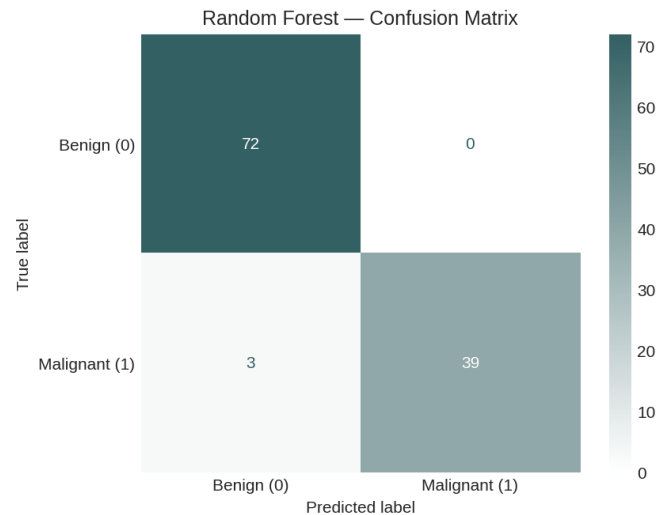


Figure 23. Random Forest confusion matrix on the test set.

The feature importance plot in Figure 24 ranks features by their mean decrease in impurity across all trees. Worst area dominates by a large margin, followed by worst concave points, worst radius, worst perimeter, and mean concave points. Once again, all size and shape features, consistent with the EDA and PCA findings. Interestingly, worst texture, which was the single most influential feature in Logistic Regression, ranks much lower here. This discrepancy reflects a fundamental difference between the two models: Logistic Regression assigns a single global weight to each feature, while Random Forest measures how useful a feature is across many different local splits. In combination with other features, worst area seems to contribute more consistently to reducing impurity than worst texture does.

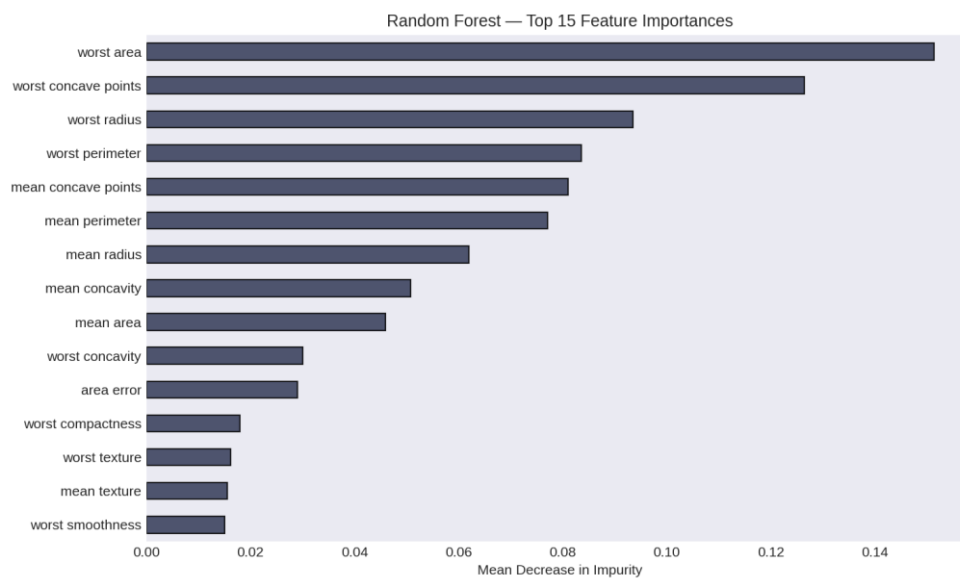


Figure 24. Random Forest — top 15 feature importances (mean decrease in impurity).

6.6 Cost-Sensitive Improvement

The baseline models were trained with equal misclassification costs for both classes. However, as established in Section 6.1, a false negative carries a far higher clinical cost than a false positive. Cost-sensitive learning addresses this directly by setting `class_weight='balanced'`, which automatically adjusts the penalty for misclassifying each class inversely proportionally to its frequency. Therefore, the malignant class receives a higher penalty since it is the minority class. This pushes the decision boundary to be more cautious, flagging more samples as malignant to avoid missing true cancer cases. Logistic Regression, Linear SVM, and Random Forest were all retrained with this setting.

k-Nearest Neighbors was excluded from the cost-sensitive analysis, as it was the worst performing model in the baseline evaluation and does not support class weighting directly. Resources were therefore focused on the three models that showed stronger potential.

Metric	Logistic Regression (balanced)	Linear SVM (balanced)	Random Forest (balanced)
Train Accuracy	0.9824	0.9802	1.0000
Test Accuracy	0.9825	0.9912	0.9737
Precision	0.9762	1.0000	1.0000
Recall	0.9762	0.9762	0.9286
F1-Score	0.9762	0.9880	0.9630
AUC	0.9977	0.9983	0.9965

Table 6. Cost-sensitive results on the test set for all three models.

Logistic Regression improves recall from 0.9524 to 0.9762, meaning one additional malignant case is correctly identified compared to the baseline. The trade-off is that precision drops from 1.000 to 0.9762, meaning that one benign case is now incorrectly flagged as malignant. The best C increased from 0.034 to 0.234, indicating that with balanced weights the model requires less regularization to generalize well. The train-test gap is negligible (0.9824 vs. 0.9825), confirming no overfitting.

Linear SVM shows the most dramatic improvement of all three models. Recall improves from 0.8810 to 0.9762 and precision remains at 1.000, meaning the model now achieves both perfect precision and high recall simultaneously. This is explained by the best C, dropping dramatically from 29.76 to 0.002. Balanced weighting completely changed the optimization landscape, selecting a much stronger regularization that eliminated the overfitting problem observed in the baseline SVM. The train-test gap disappears (0.9802 vs. 0.9912), and with F1 = 0.9880, this is the strongest performing model overall.

Random Forest shows almost no improvement over the baseline. Recall, precision, F1 remain identical, though AUC improves slightly from 0.9929 to 0.9965. This is largely expected: Random Forest is already inherently robust to class imbalance due to its bootstrapping mechanism, which naturally exposes each tree to a balanced mix of samples. Applying balanced class weights therefore has minimal effect on the final predictions, though it does marginally improve the model's overall discriminative ability as reflected by the AUC.

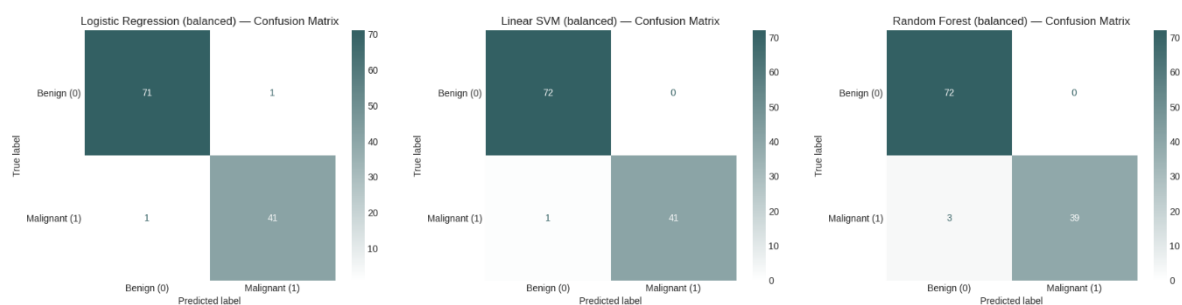


Figure 25. Confusion matrices for cost-sensitive Logistic Regression (left), Linear SVM (middle), and Random Forest (right).

6.7 Supervised Summary

Of the seven models evaluated, three met both success criteria: Logistic Regression, cost-sensitive Logistic Regression, and cost-sensitive SVM, with the latter achieving the highest recall of 0.9762 alongside a near-perfect F1 of 0.9880. The complete performance comparison is shown in Table 7 and the ROC curves for all models are shown in Figure 26.

Model	Train Acc	Test Acc	Precision	Recall	F1	AUC
Logistic Regression	0.9780	0.9825	1.0000	0.9524	0.9756	0.9980
Linear SVM	0.9956	0.9386	0.9487	0.8810	0.9136	0.9679
kNN (best k)	0.9846	0.9386	0.9730	0.8571	0.9114	0.9825
Random Forest	1.0000	0.9737	1.0000	0.9286	0.9630	0.9929
LR (balanced)	0.9824	0.9825	0.9762	0.9762	0.9762	0.9977
SVM (balanced)	0.9802	0.9912	1.0000	0.9762	0.9880	0.9983
Random Forest (balanced)	1.0000	0.9737	1.0000	0.9286	0.9630	0.9965

Table 7. Full supervised model comparison on the test set

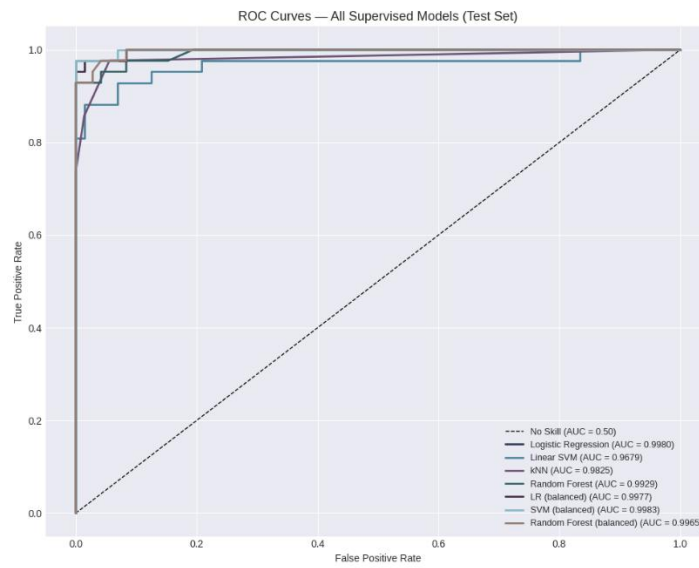


Figure 26. ROC curves for all supervised models on the test set.

Balanced Linear SVM is the strongest overall model, achieving the highest F1 (0.9880), the highest test accuracy (0.9912), perfect precision (1.000), and the highest AUC (0.9983). Crucially, it also achieves the best recall (0.9762) jointly with Balanced LR meaning only 1 malignant case is missed out of 42. The dramatic improvement of Balanced SVM over its baseline version highlights how sensitive Linear SVM is to regularization and class weighting: the best C dropped from 29.76 to 0.002, eliminating overfitting entirely.

However, raw performance on a single test set is not the only criterion for model selection in a clinical context. Balanced Logistic Regression is the recommended alternative if interpretability and clinical deployability are priorities. Its coefficients map directly to features, it produces calibrated probability scores that allow threshold tuning, and its train-test gap is negligible. kNN performs the weakest on the primary metrics, consistent with the curse of dimensionality discussed in Section 6.4. Random Forest is robust and strong, but cost-sensitive weighting brings no improvement beyond a marginal AUC gain, confirming that it is already naturally resistant to class imbalance.

7. Connection Between Unsupervised and Supervised Models

The two parts of this report, (1) unsupervised exploration and (2) supervised classification, were designed to complement each other. This section explicitly connects them by answering four questions: Does PCA help supervised classification? Do the clusters discovered in Section 5 correspond to the true diagnosis? Does adding cluster membership as a feature improve prediction? And do the unsupervised structures explain where supervised classifiers fail? Together, these questions test whether the geometric structure discovered without labels has genuine predictive meaning.

7.1 Does PCA Help Prediction?

Balanced Logistic Regression was selected for this analysis as it was the recommended model at the end of Section 6 for its interpretability and clinical deployability. It is also particularly well suited for combination with PCA: Logistic Regression is a linear model that relies on finding a linear decision boundary, and PCA's linear projections preserve the global variance structure that this boundary depends on. Additionally, PCA produces uncorrelated components, which helps Logistic Regression's regularization work more effectively. Correlated features can cause unstable and hard to interpret coefficient estimates, while uncorrelated PCA components avoid this issue entirely. Balanced Logistic Regression was therefore trained on PCA-reduced features with 2, 5, 7, and 10 components, and compared against the full 30-feature raw representation.

Features	F1	Recall
PCA (2 components)	0.9762	0.9762
PCA (5 components)	0.9639	0.9524
PCA (7 components)	0.9524	0.9524
PCA (10 components)	0.9639	0.9524
Raw (30 features)	0.9762	0.9762

Table 8. Logistic Regression: raw features vs. PCA components.

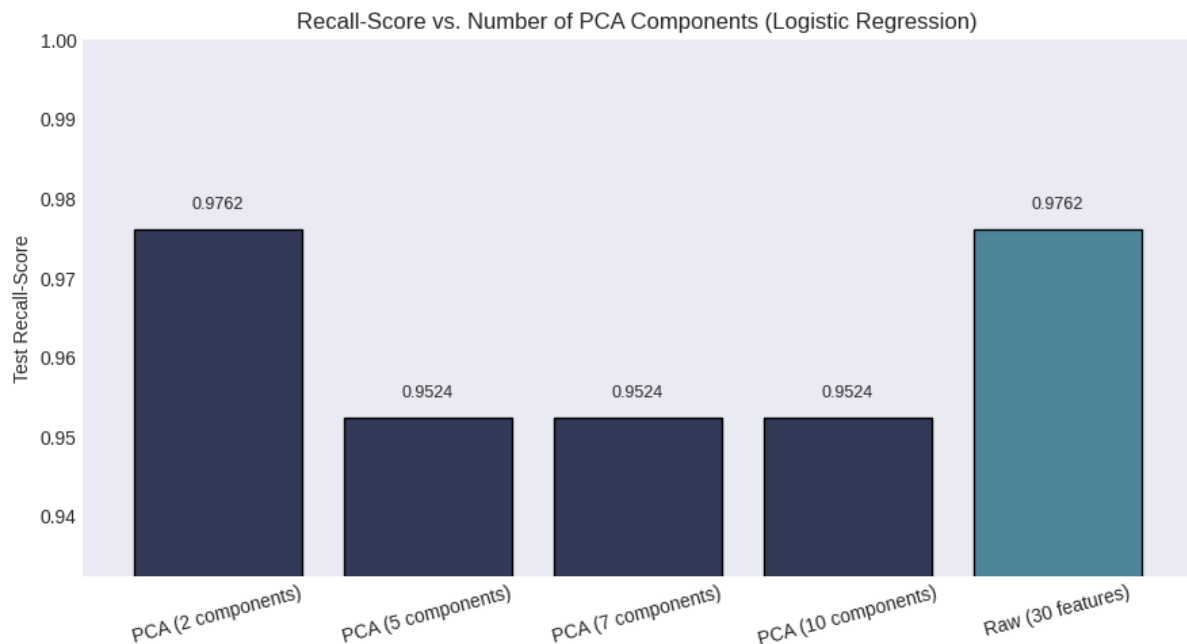


Figure 27. Recall vs. number of PCA components for Logistic Regression (balanced). Raw features shown for comparison.

The results reveal a striking finding: PCA, with just 2 components, already matches raw feature performance, achieving the same recall (0.9762) and F1 (0.9762) as the full 30-feature set. This directly

confirms what the EDA and loading analysis suggested. PC1 alone, driven by size and shape features, acts as a near-complete separating axis between benign and malignant cases, capturing so much of the discriminative signal that even a 2D projection loses nothing in terms of predictive performance.

Performance dips in the intermediate range: recall drops to 0.9524 at 5 and 7 components and stays there through 10 components, while F1 dips further to 0.9524 at 7 components before partially recovering to 0.9639 at 10 components. This non-monotonic pattern is somewhat counterintuitive, since adding more components should in principle only give the model more information to work with. A likely explanation is that components 3–10 introduce dimensions capturing variation that is largely unrelated to malignancy, such as subtle texture noise, fine boundary complexity, and smoothness fluctuations, which both classes share to a similar degree, effectively adding noise to the decision boundary before enough genuinely discriminative information accumulates at higher component counts.

The conclusion is therefore nuanced. Raw features remain the preferred choice when all 30 measurements are available, since they match or exceed PCA performance. However, the fact that 2 components already achieve full recall is a meaningful validation of the data's geometric structure: the biological signal is so concentrated along PC1 that extreme dimensionality reduction carries no clinical cost in this dataset. If imaging constraints or cost limitations restrict access to the full feature set, PCA with 2 components offers a surprisingly strong alternative with no loss in malignant case detection.

It should also be noted that this analysis was conducted exclusively with Balanced Logistic Regression, and other models may respond differently to PCA preprocessing, therefore these findings should not be generalized beyond this specific model without further testing.

7.2 Do Clusters Correspond to True Labels?

This question was already partially answered in the outlook of Section 5.5, where labels were revealed post-hoc and the answer is clearly yes. All three clustering methods recover a structure that strongly aligns with the true diagnosis without ever observing the labels. GMM achieves the highest ARI (0.7524) and recall (0.9294) on the baseline 30D data, meaning it correctly identifies over 93% of malignant cases purely from geometric structure. K-Means and Hierarchical follow with ARI values of 0.6622 and 0.6212 respectively.

The connection to the supervised results is important. The fact that unsupervised methods can recover the diagnosis with recall above 0.92 confirms that the 30 FNA measurements carry a strong and coherent biological signal. This is not a trivial finding: it means the malignant and benign classes are not just statistically separable, but geometrically distinct in the original feature space. This directly explains why supervised classifiers perform so strongly on this dataset. The signal is clear enough that even label-free geometry captures most of it. It also provides biological validation for the FNA measurements themselves: the nuclear morphology features introduced by Street et al. (1993) genuinely reflect underlying differences between malignant and benign tissue at the cellular level.

7.3 Does Cluster Membership Improve Prediction?

To answer this question, the K-Means cluster assignment for each sample was added as an extra 31st feature to the original 30 features, and Balanced Logistic Regression was retrained on this augmented dataset. K-Means was selected for this analysis as it was the recommended clustering method in Section 5.4 and produces hard cluster assignments that translate naturally into a single additional feature. GMM's soft probabilities could also be used but would require a different experimental setup. The idea is to test whether knowing which cluster a sample belongs to gives the model any additional information beyond what the 30 original features already provide.

The results show that it does not. F1 decreases from 0.9762 to 0.9639, and recall drops from 0.9762 to 0.9524. The cluster feature not only fails to improve performance but marginally hurts it, likely because it introduces a redundant and noisy signal that the regularization must work around.

This is an informative finding rather than a disappointing one. It confirms that the supervised model already extracts everything the clustering captures, since both are derived from the same 30 features. The cluster label is essentially a compressed and lossy version of information the model can already access directly from the raw features. In other words, K-Means clustering and Logistic Regression are discovering the same underlying structure in the data, just from different angles. This further validates the biological coherence of the 30 FNA measurements: the signal is strong and consistent enough that multiple fundamentally different methods converge on the same answer.

7.4 Do Unsupervised Methods Explain Model Errors?

To answer this question, the misclassified samples from the Balanced Logistic Regression and GMM are visualized in PCA space and compared. The GMM used here was refitted on training data only and evaluated on the test set, consistent with the rest of the methodology.

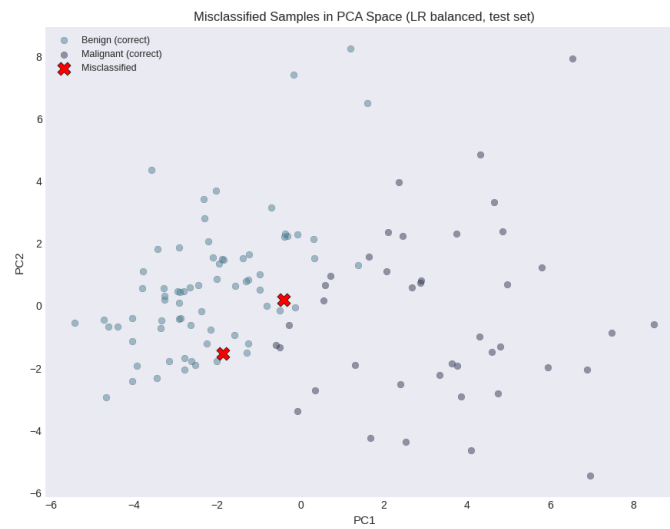


Figure 28. Misclassified samples in PCA space — Logistic Regression (balanced), test set. Red crosses indicate misclassified samples.



Figure 29. Misclassified samples in PCA space — GMM, test set. Red crosses indicate misclassified samples.

Balanced Logistic Regression misclassifies only 2 samples out of 114, both located in the central overlap zone along PC1, where benign and malignant samples mix most densely. This is precisely the region identified by GMM as geometrically ambiguous throughout Section 5, the boundary where soft probabilities approach 0.5 and the model is genuinely uncertain about cluster assignment. The fact that the two hardest cases for the supervised classifier fall exactly in the region of highest unsupervised

uncertainty is not a coincidence. It confirms that the difficulty is inherent to the data geometry itself, not a limitation of any specific method.

GMM misclassifies substantially more samples, as expected for an unsupervised method without access to labels. However, the misclassified samples are again concentrated in the same central overlap zone around $PC1 = 0$, with only a few scattered further away. Importantly, some samples that GMM struggles with, particularly those at higher $PC2$ values, are correctly classified by LR. This shows that access to labels allows the supervised model to resolve ambiguities that unsupervised geometry alone cannot, by learning the precise decision boundary rather than relying purely on cluster structure.

These findings confirm that the errors made by both supervised and unsupervised methods are not random. They are concentrated in a geometrically well-defined region of feature space that was identified as ambiguous throughout Section 5, long before any labels were used.

Together, the four questions answered in this section paint a coherent picture. The 30 FNA measurements carry a strong and genuine biological signal that is strong enough that unsupervised methods recover the diagnosis with high recall and that supervised classifiers achieve near-perfect performance. Where classifiers do fail, the errors are not random but concentrated in the geometrically ambiguous overlap zone identified by unsupervised methods long before any labels were used. This consistency across methods and perspectives is the strongest possible validation of the biological meaningfulness of the FNA measurements introduced by Street et al. (1993).

8. Reflection

8.1 What Worked Well

The PCA separation was remarkably clean: even in 2D, the two clusters were visually separable without using any labels, confirming that the 30 FNA features carry a strong and coherent biological signal. PC1 alone, driven by size and shape measurements, acts as a near-complete separating axis between benign and malignant cases. This was consistent with the EDA findings and validated post-hoc by the label-based verification in Section 5.5.

The best supervised models exceeded the success criteria: Balanced Linear SVM (recall = 0.9762, F1 = 0.9880) and Balanced Logistic Regression (recall = 0.9762, F1 = 0.9762), with the gap over their baseline counterparts highlighting the value of cost-sensitive weighting. Cost-sensitive weighting proved highly effective, particularly for Linear SVM, where it simultaneously improved recall and eliminated overfitting by driving the best C from 29.76 to 0.002.

The connection between unsupervised and supervised findings was the most satisfying outcome of the project. Both methods struggled with the same samples, concentrated in the same geometrically ambiguous overlap zone along PC1, a region identified without any label information in Section 5, long before supervised training began.

8.2 What Did Not Work as Well

kNN performed below all linear models on the primary metrics despite careful tuning, achieving recall of 0.8571 compared to 0.9762 for the best model. In a 30-dimensional feature space, distance-based methods suffer from the curse of dimensionality. This disadvantage could not be resolved by standardization alone.

Adding the K-Means cluster assignment as an extra 31st feature produced no improvement. In fact, there was a slight decrease in F1 (0.9762 vs. 0.9639). The cluster label is entirely redundant given the full 30-feature set: the supervised model already extracts all the relevant signal directly from the raw features, and the additional column only introduces noise.

Silhouette scores for t-SNE representations appeared artificially high compared to the baseline 30D scores. However, this is a known limitation of t-SNE. It manufactures tight, well-separated clusters by design, making silhouette scores in t-SNE space not directly comparable to those in the original feature space. This is why the baseline 30D silhouette was treated as the most honest measure of cluster quality throughout Section 5.

8.3 What Surprised Us

The degree of unsupervised structure was striking. All three clustering methods achieved positive silhouette scores on the raw 30D features without any label information, and the post-hoc label verification revealed that GMM alone correctly identified over 93% of malignant cases purely from geometry. This suggests that malignant and benign tumors are not just statistically separable but genuinely distinct at the level of nuclear morphology, which is a meaningful biological result beyond just a machine learning one.

The dramatic improvement of Linear SVM with balanced weighting was also unexpected. The baseline SVM was the weakest linear model due to severe overfitting (train accuracy 0.9956, test accuracy 0.9386). Simply adding `class_weight='balanced'` transformed it into the strongest overall model, eliminating the overfitting problem by selecting a much stronger regularization ($C = 0.002$ vs. 29.76).

8.4 Most Convincing Methods

On the supervised side, Balanced Linear SVM achieved the best overall performance with recall = 0.9762 and F1 = 0.9880, which are the strongest results across all metrics. Nonetheless, Balanced Logistic Regression is the recommended model for clinical deployment: its feature coefficients are

directly interpretable, it produces calibrated probability scores that allow threshold tuning, and its train-test gap is negligible.

On the unsupervised side, K-Means achieved the highest silhouette score on the baseline 30D data (0.3432), making it the strongest method by the primary internal evaluation metric. However, GMM was the most convincing method overall: despite a slightly lower silhouette score (0.3142), it achieved the highest ARI (0.7524) and recall (0.9294) in the post-hoc label verification, and its soft probability assignments uniquely identified the geometrically ambiguous boundary samples that proved hardest for supervised classifiers to handle. The fact that GMM's uncertainty map and the supervised classifier's errors overlap so closely is the most compelling methodological finding of this project.

8.5 Limitations

This analysis has several important limitations that should be acknowledged.

First, all evaluation is performed on a held-out portion of the same dataset, so there is no external validation on independent data, and generalization to other FNA datasets, imaging protocols, or patient populations cannot be guaranteed.

Second, the Wisconsin Breast Cancer dataset is a clean and well-structured benchmark with no missing values and a strong biological signal. Real-world clinical data is rarely this cooperative, and the strong results achieved here should be interpreted as an upper bound on real-world performance rather than a deployment guarantee.

Third, the 30 features are computed from digitized images, meaning any errors or variability in the image processing pipeline propagate invisibly into the analysis.

Finally, with only 569 samples, cross-validation estimates may be sensitive to the specific train-test split, and larger datasets would yield more stable and reliable performance estimates.

9. Ways to Improve

The models developed in this project already exceed the success criteria; however, the following improvements could meaningfully enhance performance, generalizability, or clinical deployability.

9.1 Non-Linear Models

All classifiers in this project used linear decision boundaries or shallow ensembles. While PCA showed near-linear separability, the overlap zone identified in Section 7.4 suggests that a small amount of non-linear structure remains. Kernel SVM would allow the model to learn non-linear boundaries in the original feature space at no additional data cost, potentially resolving some of the ambiguous cases in the overlap zone. Moreover, gradient boosting methods such as XGBoost could also be explored as stronger ensemble alternatives to Random Forest.

9.2 Threshold Optimization

All models were evaluated at the default decision threshold of 0.5. The cost-sensitive weighting applied in Section 6.6 already partially addresses this by penalizing false negatives more heavily during training, effectively shifting the decision boundary toward greater caution. However, a complementary and more explicit approach is direct threshold optimization via precision-recall curve analysis, which would allow selection of a specific operating point without any retraining, which matches the acceptable false-negative rate for a given clinical protocol. This gives the clinician direct control over the precision-recall trade-off at deployment time, independently of the training process.

10. Conclusion

This project investigated the Wisconsin Breast Cancer dataset from both unsupervised and supervised perspectives. The unsupervised analysis confirmed that the 30 FNA nucleus measurements carry a strong and coherent biological signal: all three clustering methods applied produced well-separated clusters with positive silhouette scores without ever observing the diagnosis labels, with K-Means achieving the highest baseline silhouette (0.3432). PCA reduced the 30 features to 7 components capturing 90% of the total variance, and even a 2D projection revealed clear cluster separation along PC1. When labels were revealed post-hoc, GMM correctly identified 93% of malignant cases purely from geometric structure, confirming that the discovered clusters correspond closely to clinical reality.

On the supervised side, several classifiers outperformed the baseline. Balanced Linear SVM achieved the strongest overall performance with recall = 0.9762 and F1 = 0.9880, while Balanced Logistic Regression matched its recall and offers the additional advantages of interpretability and calibrated probability outputs. Cost-sensitive weighting proved highly effective, most dramatically for Linear SVM, where it simultaneously improved recall from 0.8810 to 0.9762 and eliminated overfitting entirely.

The connection between unsupervised and supervised learning methods was most vividly confirmed by the spatial distribution of misclassified samples: both methods struggled with the same samples, concentrated in the geometrically ambiguous overlap zone along PC1 identified in Section 5 long before any labels were used.

The final recommendation is Balanced Logistic Regression: it achieves recall = 0.9762 and F1 = 0.9762, provides directly interpretable feature coefficients, and outputs calibrated probability scores that allow clinicians to adjust the decision threshold according to the acceptable risk level. In a clinical context where a missed malignancy can be life-threatening, this combination of strong performance, transparency, and flexibility makes it the most trustworthy choice for deployment as a diagnostic support tool.

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